

Solutions to JEE Advanced -8 | JEE 2022 | Paper-1

PHYSICS

1.(B) $\frac{dy}{dx} = 2ax$

$$\Rightarrow \tan \theta = 2ax = 2a\sqrt{\frac{y}{a}} = 2\sqrt{ay}$$

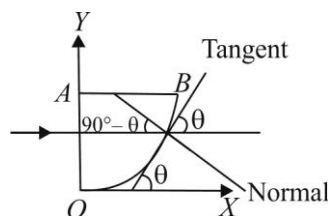
For refraction at surface OB

$$90^\circ - \theta < \theta_c$$

$$\Rightarrow \cos \theta < \sin \theta_c$$

$$\frac{1}{\sqrt{4ay+1}} < \frac{1}{\mu}$$

$$\Rightarrow y > \frac{\mu^2 - 1}{4a} \Rightarrow y_{\min} = \frac{\mu^2 - 1}{4a}$$



2.(B) Force of buoyancy on soap

$$F_B = A[x\rho_0 + (L-x)\rho_w]g$$

$$\text{Weight } W = LA\rho_s g$$

$$\text{For equilibrium } F_B = W$$

$$\Rightarrow LA\rho_s g = A[x\rho_0 + (L-x)\rho_w]g \Rightarrow L\rho_s = x\rho_0 + (L-x)\rho_w$$

$$\Rightarrow \rho_s = \frac{x}{L}(\rho_0 - \rho_w) + \rho_w \Rightarrow \frac{x}{L} = \frac{\rho_w - \rho_s}{\rho_w - \rho_0} = \frac{1000 - 800}{1000 - 300} = \frac{2}{7}$$

3.(C) For small change $dq = du + dw$

$$nC_dT = nC_vdT + 2nRdT \quad \therefore C = C_v + 2R; \text{ and } C = 4R \text{ (Given)}$$

$$4R = C_v + 2R \quad \therefore C_v = 2R$$

$$\text{Also } C_v = \frac{R}{\gamma - 1} \quad \therefore \frac{R}{\gamma - 1} = 2R \Rightarrow 2\gamma - 2 = 1 \Rightarrow \gamma = \frac{3}{2}$$

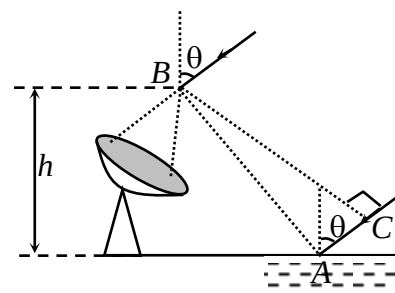
4.(D) Path difference of the two rays (one coming directly to antenna and other after reflection from water surface)

$$\Delta x = AB + AC + \frac{\lambda}{2} \Rightarrow \Delta x = \frac{h}{\cos \theta} + \frac{h}{\cos \theta} \cos 2\theta + \frac{\lambda}{2}$$

$$= \frac{h}{\cos \theta} (1 + \cos 2\theta) + \frac{\lambda}{2} \quad \text{or} \quad \Delta x = 2h \cos \theta + \frac{\lambda}{2}$$

For constructive interference $\Delta x = n\lambda$

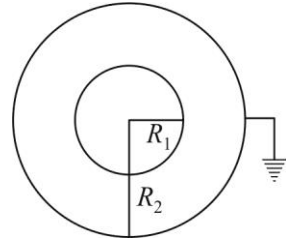
$$\Rightarrow 2h \cos \theta + \frac{\lambda}{2} = n\lambda \quad \text{or} \quad h = \frac{\left(n - \frac{1}{2}\right)\lambda}{2 \cos \theta} = \frac{\lambda}{4 \cos \theta}, \frac{3\lambda}{4 \cos \theta}, \dots \text{ etc}$$



5.(C) Given $\frac{R_2}{R_1} = \frac{n}{(n-1)}$

Then $\frac{R_1}{(R_2 - R_1)} = (n-1)$ or $\frac{R_2}{R_2 - R_1} = n$ or $\frac{R_1 R_2}{R_2 - R_1} = n R_1$

or $4\pi\epsilon_0 \left(\frac{R_1 R_2}{R_2 - R_1} \right) = 4\pi\epsilon_0 (n R_1)$ or $C_2 = n C_1$



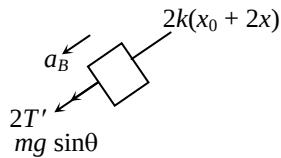
6.(B) Let elongation of spring be x_0 in equilibrium. Then,

$$2T + mg \sin \theta = 2kx_0 \quad \dots(i)$$

and $T = mg \quad \dots(ii)$

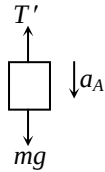
Let Block B is displaced by x down the inclination

F.B.D. of B



$$-ma_B = 2k(x_0 + 2x) - 2T' - mg \sin \theta \quad \dots(iii)$$

F.B.D. of A



$$mg - T' = ma_A$$

Also, $a_A = 2a_B$

$$T' = mg - 2ma_B \quad -ma_B = 2kx_0 + 4kx - 2mg + 4ma_B - mg \sin \theta$$

$$-ma_B = 4kx + 4ma_B; \quad a_B = -\frac{4k}{5m}x$$

$$\therefore T = 2\pi\sqrt{\frac{5m}{4k}}; \quad T = 6.28 \text{ s.}$$

7.(ACD) Centrifugal force $= \frac{dU}{dr} = kr = \frac{mv^2}{r}$

Angular momentum, $L = mvr = \frac{nh}{2\pi}$

On solving, we get $r = \sqrt{\frac{nh}{2\pi\sqrt{km}}}$ and $v = \sqrt{\frac{nh}{2\pi m} \sqrt{\frac{k}{m}}}$; $KE = \frac{1}{2}mv^2 = \frac{nh}{4\pi} \sqrt{\frac{k}{m}}$

$$U = \frac{kr^2}{2} = \frac{k}{2} \left(\frac{nh}{2\pi\sqrt{km}} \right) = \frac{nh}{4\pi} \sqrt{\frac{k}{m}}; \quad E = kE + U = \frac{nh}{2\pi} \sqrt{\frac{k}{m}}$$

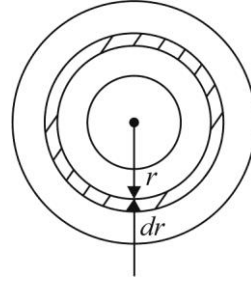
$$8.(AC) \quad C = \frac{4\pi\epsilon_0 abK}{b-a}; \quad R = \int_a^b \rho \frac{dr}{4\pi r^2} = \frac{\rho}{4\pi} \frac{(b-a)}{ab}$$

$$\tau = RC = \rho K \epsilon_0; \quad q = q_0 e^{-t/\tau}$$

$$q = q_0 e^{-\frac{t}{\rho K \epsilon_0}}; \quad t = \infty, \quad q = 0$$

Amount of heat generated

$$H = \left(\frac{1}{a} - \frac{1}{b} \right) \frac{q_0^2}{8\pi\epsilon_0 K}$$



$$9.(ABC) \quad T_1 = 2\pi\sqrt{LC_1} = \frac{4\pi}{100}$$

$$t_1 = \frac{T_1}{8}$$

Hence half of the energy is transferred to the inductor

$$T_2 = 2\pi\sqrt{LC_2} = \frac{2\pi}{100}; \quad t_2 = t - t_1 = \frac{\pi}{200} = \frac{T_2}{4}$$

Hence all the energy of the inductor is transferred to the 2nd capacitor

$$\frac{1}{2} \left(\frac{1}{2} (400) (\sqrt{2})^2 \right) = \frac{1}{2} 100 (V_2)^2 \Rightarrow V_2 = 2V$$

10.(ABD)

$$(A) \quad v = \sqrt{\frac{T}{\mu}} \Rightarrow T = \mu v^2 = (0.5 \times 10^{-1}) (0.2)^2 = 2 \times 10^{-3} N$$

$$(B) \quad \text{time taken} = \text{time after which pulse will reach the point after reflecting from the free end} \\ = \frac{20+20}{20} = 2s$$

$$(D) \quad \text{time taken} = \text{time after which pulse will reach the point after 2 reflections from fixed end} \\ = \frac{20+30+30+30+10}{20} = 6s$$

11.(BC)

$$\frac{dQ}{dt} = eS \times A = ms \frac{dT}{dt} \quad (\because \delta = \sigma T^4)$$

$$\Rightarrow \frac{dT}{dt} \text{ is more in spirit}$$

In steady state

$$eSA = e \times \sigma (T_4 - T_0^4)$$

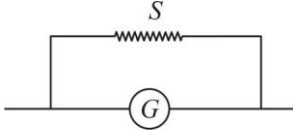
$$\Rightarrow T \text{ is same in both.}$$

12.(BC)

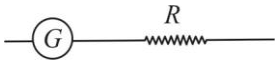
$$\frac{v_A}{v_B} = \sqrt{\frac{\gamma_A M_B}{\gamma_B M_A}} = \sqrt{\frac{7 \times 3}{5 \times 5} \times \frac{4}{2}} = \sqrt{\frac{42}{25}}; \quad f_A = \frac{3V_A}{2\ell_A} \text{ (second overtone of A)}$$

$$\frac{3v_A}{2\ell_A} = \frac{3v_B}{4\ell_B} \Rightarrow \frac{\ell_A}{\ell_B} = \frac{2v_A}{v_B} = \sqrt{\frac{168}{25}}; \quad f_B = \frac{3V_B}{4\ell_B} \text{ (first overtone of B)}$$

1.(9), 2.(10)



$$I_0 + I_0 \frac{G}{S} = nI_0 \Rightarrow G = (n-1)S \Rightarrow G = (n-1)3$$



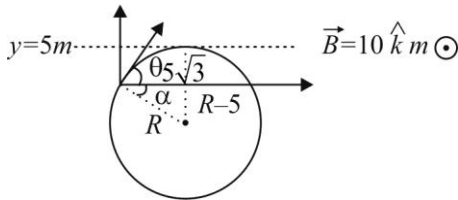
$$I_0(G+R) = nI_0G \Rightarrow G+R = nG \Rightarrow G+27 = n(G)$$

Solving the equations, we get

$$n = 4, \quad G = 9\Omega$$

$$I_0^2 G = 9 \times 10^{-4} \Rightarrow I_0 = 10^{-2} A = 10mA$$

3.(60), 4.(2)



$$(5\sqrt{3})^2 + (R-5)^2 = R^2 \Rightarrow R = 10m$$

$$\tan \alpha = \frac{5}{5\sqrt{3}} = \frac{1}{\sqrt{3}} \Rightarrow \alpha = 30^\circ$$

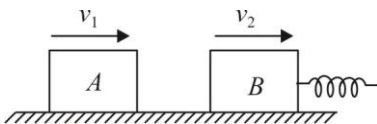
$$\Rightarrow \theta = 90 - \alpha = 60^\circ$$

$$R = \frac{mv}{qB} \Rightarrow v = \frac{qBR}{m} = \frac{10^{-6} \times 10 \times 10}{5 \times 10^{-5}} = 2m/s$$

5.(10)

$$T = 2\pi\sqrt{\frac{m}{k}} \Rightarrow 0.04 = 4 \times 10 \times \left(\frac{m}{1000}\right) \Rightarrow m = 1kg$$

Immediately after the collision, suppose velocities of the blocks are v_1 and v_2 as shown



$$\frac{1}{2} \text{ (velocity of approach)} = \text{velocity of separation} \Rightarrow 5 = v_2 - v_1 \quad \dots(i)$$

Using principle of conservation momentum for the collision

$$2 = 0.2v_1 + v_2 \Rightarrow 10 = v_1 + 5v_2 \quad \dots(ii)$$

On solving $v_2 = 2.5 \text{ m/s}$ and $v_1 = -2.5 \text{ m/s}$.

Hence block A moves leftward after the collision with speed 2.5 m/s .

And the block B moves towards right with speed 2.5 m/s .

The maximum velocity of $B = 2.5 = \omega A \Rightarrow A = v \sqrt{\frac{m}{k}} = 2.5 \sqrt{\frac{1}{1000}} \text{ m} = 2.5 \sqrt{10} \text{ cm}$

6.(2.5)

$$\text{Time of flight} = \sqrt{\frac{2h}{g}} = 1 \text{ s} \Rightarrow d = (2.5 \text{ m/s}) \times 1 \text{ s} = 2.5 \text{ m}$$

CHEMISTRY

1.(B)

2.(A) Silica- SiO_2 - oxide of p-block $\rightarrow$ acidic

Lime, dolomite = basic

Bone ash = Neutral

3.(B) at room temp. ortho = 75%

Para = 25%

4.(A) $4\text{KO}_2 + 2\text{CO}_2 \longrightarrow 2\text{K}_2\text{CO}_3 + 3\text{O}_2$

5.(C) $\text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow 4\text{K}^{-1} + [\text{Fe}(\text{CN})_6]^{-4}$

$a - a\alpha \qquad 4a\alpha \qquad a\alpha$

$$i = 1 + 4\alpha = 1 + 4 \times 0.1$$

$$= 1.4$$

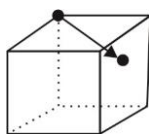
$$\Delta T_b = iK_b m = 1.4 \times 0.513 \times 0.1$$

$$= 0.072$$

$$T_b = 100 + 0.072$$

$$= 100.072^\circ\text{C}$$

6.(C)

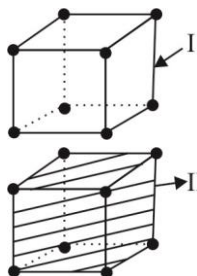
Number = 6×4

$$\text{distance} = a\sqrt{1.5}$$

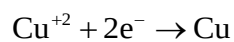
7.(ABC)

Due to better overlapping in p-orbitals of C-C bonds are more stable and have shorter length.

8.(A)

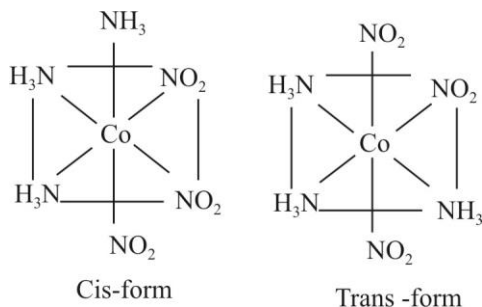


9.(AC) $\text{mole}_{\text{Cu}} = \frac{0.635}{63.5} = 0.01$

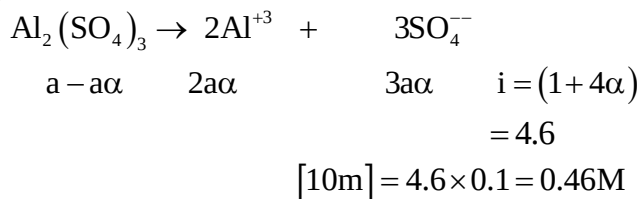


Req mole of electron = 0.02

10.(B)



11.(CD)



$$\text{BaCl}_2 [\text{ions}] = 2 \times 0.1 = 0.2M$$

$$\text{Na}_2\text{SO}_4 [\text{ions}] = 3 \times 0.05 = 0.15M$$

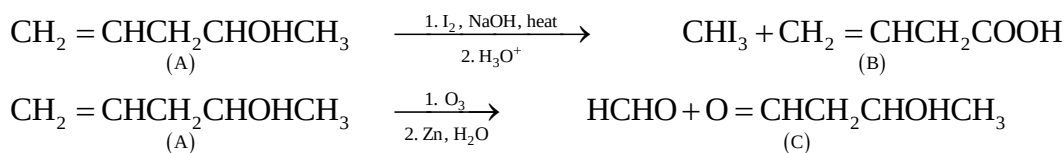
$$\text{urea} = 0.15M$$

$$\text{mole } \uparrow \Rightarrow \text{B.P } \uparrow \Rightarrow \text{B.P : } A > B > C = D$$

12.(ABCD)

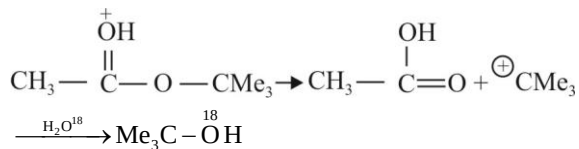
1.(2)

2.(1.3) The corresponding hydrocarbon of the compound (A) is C_5H_{10} and the corresponding alkane is C_5H_{12} . So, the index of hydrogen deficiency of (A) is 1. Since (A) on ozonolysis gives one mole of HCHO and an aldehyde (C), the compound is acyclic and contains a $\text{CH}_2 = \text{CH} -$ group. The compound (A) does not form 2,4-DNP derivative. So it is neither an aldehyde, nor a ketone. Since it gives iodoform reaction, it must contain a $\text{CH}_3\text{CHOH} -$ group. As (A) is acyclic with molecular formula $\text{C}_5\text{H}_{10}\text{O}$ and contains a $\text{CH}_2 = \text{CH} -$ group and a $\text{CH}_3\text{CHOH} -$ group, the structure of (A) must be $\text{CH}_2 = \text{CHCH}_2\text{CHOHCH}_3$. The structures of the compounds (B) to (C) can be established on the basis of the reactions carried out on (A) as follows.



3.(4) Acidic hydrolysis (unimolecular) involves formation of carbocation (alkyl cleavage) in the rate determining step. As the same carbocation forms in the rate determining step, the rate of reaction remains almost same for all.

4.(60) Acidic strength : $\text{CH}_3\text{COOH} > \text{Me}_3\text{C}-\text{OH}^{18}$



$$5.(75) \quad K = Ae^{-E_a/RT}$$

$$\ln \frac{K_1}{K_2} = \frac{E_a}{R} \left[\frac{1}{T_2} - \frac{1}{T_1} \right]$$

$$\ln \frac{1}{1.1} = \frac{E_a}{8.3} \left[\frac{1}{301} - \frac{1}{300} \right]$$

$$\ln 11 - \ln 10 = \frac{E_a}{8.3} \times \frac{1}{301 \times 300}$$

$$2.4 - 2.3 = \frac{E_a}{8.3 \times 301 \times 300}$$

$$E_a = 75 \text{ KJ / mole}$$

$$6.(66.67)$$

$$\ln \frac{K_1}{K_2} = \frac{E_a}{R} \left[\frac{1}{T_2} - \frac{1}{T_1} \right]$$

$$\ln \frac{K_1}{K_2} = \frac{8700}{2} \left[\frac{1}{300} - \frac{1}{290} \right]$$

$$\ln \frac{K_1}{K_2} = \frac{8700}{2} \times \frac{(-10)}{300 \times 290}$$

$$\ln \frac{K_1}{K_2} = -0.5 \quad \log \frac{K_2}{K_1} = 0.22$$

$$\log \frac{K_2}{K_1} = 1 - 0.78$$

$$\log \frac{K_2}{K_1} = \log 10 - \log 6$$

$$\frac{K_2}{K_1} = \frac{10}{6}$$

$$\frac{K_2}{K_1} = 1.66667$$

Increases in rate = 66.67%

MATHEMATICS

1.(D) Equation of chord of contact form $P(x_1, y_1)$

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \quad \dots(i)$$

Similarly form $Q(x_2, y_2)$, $\frac{xx_2}{a^2} - \frac{yy_2}{b^2} = 1 \quad \dots(ii)$

$$\therefore \text{Product of slopes} = -1 \Rightarrow \frac{x_1 x_2}{y_1 y_2} = -\frac{a^4}{b^4}$$

2.(B) $\frac{1-f(x)}{x}, \frac{1+f(x)}{x} \geq 0 \quad \forall x \in [1, 3]$

$$\int_1^2 \frac{1-f(x)}{x} dx \geq \int_1^2 \frac{1-f(x)}{2} dx \quad \dots (i)$$

$$\int_2^3 \frac{1+f(x)}{x} dx \leq \int_2^3 \frac{1+f(x)}{2} dx$$

$$\Rightarrow -\int_2^3 \frac{1+f(x)}{x} dx \geq -\int_2^3 \frac{1+f(x)}{2} dx \quad \dots (ii)$$

Adding equation (i) and (ii), we get $\ln 2 - \ln\left(\frac{3}{2}\right) \geq \int_1^3 \frac{f(x)}{x} dx$

3.(B) $f(x) = \left(x^3 + \frac{1}{x^3}\right) - 6\left(x^2 + \frac{1}{x^2}\right) + 12\left(x + \frac{1}{x}\right) + a \geq 0 \quad \forall x > 0$ and

$$f(x) = \left(x^3 + \frac{1}{x^3}\right) - 6\left(x^2 + \frac{1}{x^2}\right) + 12\left(x + \frac{1}{x}\right) + a \leq 0 \quad \forall x < 0$$

Let $x + \frac{1}{x} = t$

So, $g(t) = t^3 - 6t^2 + 9t + 12 + a \geq 0; t \geq 2$

$$g'(t) = 3(t-1)(t-3) \Rightarrow g(3) \geq 0 \text{ and } g(t) \leq 0 \quad \forall t \leq -2 \Rightarrow -12 \leq a \leq 38$$

4.(B) Let $g(x) = f(x) - x^2 \Rightarrow g(x)$ has atleast 3 real roots which are $x = 1, 2, 3$

By lagrange mean value theorem (LMVT)

$$\Rightarrow g'(x) \text{ has atleast 2 real roots in } x \in (1, 3) \Rightarrow g''(x) \text{ has atleast 1 real roots in } x \in (1, 3)$$

$$\Rightarrow f''(x) - 2 = 0 \text{ for atleast 1 real roots in } x \in (1, 3) \Rightarrow f''(x) = 2 \text{ for atleast one } x \in (1, 3)$$

5.(A) Let $I(a) = \int_0^\infty \frac{\tan^{-1} ax - \tan^{-1} x}{x} dx$

$$I'(a) = \frac{1}{a^2} \int_0^{\infty} \frac{dx}{x^2 + \frac{1}{a^2}} = \frac{\pi}{2a}$$

$$\therefore I = \frac{\pi}{2} \ln a + c$$

$$\text{So, } I(a) = \frac{\pi}{2} \ln a$$

6.(D) Case – I : Let $x \leq 0$ then $2.3^x \leq 2$ and $\frac{4x^2 + x + 2}{x^2 + x + 1} \geq 2 \Rightarrow x \leq 0$ or $x \geq \frac{1}{2}$. So, $x \leq 0$

Case –II: Let $x > 0$, we prove that $\frac{4x^2 + x + 2}{x^2 + x + 1} < 2.3^x$

Assume the opposite i.e. $\frac{4x^2 + x + 2}{x^2 + x + 1} \geq 2.3^x \quad \therefore \quad \frac{4x^2 + x + 2}{x^2 + x + 1} > 2.3^0 = 2 \Rightarrow x < 0$ or $x > \frac{1}{2}$

Since, $x > 0$. So, $x > \frac{1}{2}$. Hence, $\frac{4x^2 + x + 2}{x^2 + x + 1} \geq 2.3^x > 2\sqrt{3} > 3$

$$\Rightarrow x^2 - 2x - 1 > 0 \Rightarrow x \in (1 + \sqrt{2}, \infty)$$

$$\text{Thus, } \frac{4x^2 + x + 2}{x^2 + x + 1} \geq 2.3^x > 23^{1+\sqrt{2}} > 2.3^2 = 18$$

But $\frac{4x^2 + x + 2}{x^2 + x + 1} < 4$ for any $x > 0$, we get a contradiction

So, domain is R

7.(BC) (A) $f(x) = \{x\} \sin^2 \pi x$ differentiable $\forall x \in R$

(B) not differentiable at $x = \pi$

(C) not differentiable at $x = 0, 2, 6, 4$

(D) not differentiable at all integers

8.(ABD) Let $k(x) = f(x)^2 + f'(x)^2$ using MVT, $|f'(a)| = \left| \frac{f(2) - f(0)}{2} \right| \leq 1: a \in (0, 2)$

$$\text{So, } k(a) \leq 2$$

$$\text{Similarly; } k(b) \leq 2, b \in (-2, 0)$$

$\therefore k(0) = 4$ so $k(x)$ has a maximum at some interior point of $(-2, 2)$.

Let that point be x_0 then certainly $k(x_0) \geq k(0) = 4 \Rightarrow k'(x_0) = 0$

$$\Rightarrow f'(x_0)(f''(x_0) + f'(x_0)) = 0: \text{ clearly, } f'(x_0) \neq 0$$

9.(AC) $h(f(x))$ is continuous $\forall x \in R$ and not differentiable at $x = -1, 1, 2$

10.(BC) $f(x)$ has maximum value 2 when $x = -5$ and $g(x)$ has maximum value 3 when $x = -5$

11.(ABC)

$$(B) \quad \int_0^1 f(t)dt = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} f\left(\frac{r}{n}\right) \quad \dots (1)$$

$$\int_0^1 f^2(t)dt = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} f^2\left(\frac{r}{n}\right) \quad \dots (2)$$

$$\sum_{r=1}^n \frac{1}{n} f^2\left(\frac{r}{n}\right) \geq \left(\sum_{r=1}^n \frac{1}{n} f\left(\frac{r}{n}\right) \right)^2$$

$$\text{Because } \frac{a_1^2 + a_2^2 + \dots + a_n^2}{n} \geq \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right)^2$$

With equality holding only when $a_1 = a_2 = \dots = a_n$

$$\text{Hence } \int_0^1 f^2(t)dt \geq \left[\int_0^1 f(t)dt \right]^2$$

$$(C) \quad \int_0^b h^{-1}(x)dx = \int_0^{h^{-1}(b)} yh'(y)dy$$

$$\Rightarrow \int_0^a h(x)dx + \int_0^b h^{-1}(x)dx = \int_0^a h(x)dx + yh(y)\Big|_0^{h^{-1}(b)} - \int_0^{h^{-1}(b)} h(y)dy$$

$$= \int_{h^{-1}(b)}^a h(x)dx + bh^{-1}(b) \geq h(h^{-1}(b))[a - h^{-1}(b)] + bh^{-1}(b) \geq ab$$

$$12.(ACD) \quad \lim_{x \rightarrow 0} \frac{1 - \cos nx}{1 - \cos x} = n^2$$

So, absolute value of the integrand is bounded as $x \rightarrow 0$ and hence the integral converges

$$\frac{l_{n+1} + l_{n-1}}{2} = \int_0^\pi \frac{1 - \cos nx \cdot \cos x}{1 - \cos x} = \int_0^\pi \frac{(1 - \cos nx) + \cos nx(1 - \cos x)}{1 - \cos x} = l_n$$

$$\text{So, } l_n = \frac{1}{2}(l_{n+1} + l_{n-1}); n \geq 1$$

$$\therefore l_0 = 0, l_1 = \pi, \dots, l_n = n\pi$$

1.(5) Point from which length of tangents to these circles same is radical centre

$$S_1 - S_2 = 0 \Rightarrow 4x - 4y - 4 = 0 \Rightarrow x - y - 1 = 0$$

$$S_2 - S_3 = 0 \Rightarrow -6x + 14y - 10 = 0 \Rightarrow -3x + 7y - 5 = 0$$

$$-3x + 7y - 5 = 0$$

$$3x - 3y - 3 = 0$$

$$\frac{4y - 8 = 0 \Rightarrow y = 2}{x = 3}$$

- 2.(3) If circle be drawn taking radical centre as centre and length of tangents from radical centre to any circle as radius will cut all the three circles orthogonally

$$\text{Length of tangent} = \sqrt{9+4+9+4+1} = \sqrt{S_1} = \sqrt{27}$$

3.(36) Let any symmetric matrix $B = \begin{bmatrix} a & x & y \\ x & b & z \\ y & z & c \end{bmatrix}$

$$\text{So total matrices possible} = 3! \times 3! = 36.$$

- 4.(12) (2) For homogeneous system infinite solution are possible if $|B| = 0$

$$|B| = abc + 2xyz - by^2 - cx^2 - az^2 \quad abc = xyz = 0$$

$$\text{If } a = 0 \quad z = 0 \Rightarrow |B| = 0 \quad -4 \text{ cases}$$

$$b = 0 \quad y = 0 \Rightarrow |B| = 0 \quad -4 \text{ cases}$$

$$c = 0 \quad x = 0 \Rightarrow |B| = 0 \quad -4 \text{ cases}$$

So total cases are 12.

5.(74) Identical letters are 1,1,2,2,3,3. Total words $= \frac{8!}{2! 2! 2!} = 5040$

Number of numbers in which all identical digits are together $= 5!$

$$\text{Number of numbers in which only exactly two pair of identical digits appear together} = \left(\frac{6!}{2!} - 5! \right) \times 3 = 720$$

Number of words in which only one pair identical digits appear together (1,1) together

$$= \left(\frac{7!}{2! 2!} - \{240 + 240 + 120\} \right) \times 3 = 1980 \text{ Now number of words in which no two identical digits appear}$$

$$\text{together} = 5040 - (1980 + 720 + 120) = 2220 \text{ Probability} = \frac{2220}{5040} = \frac{37}{84}$$

6.(9) Number of numbers in which only exactly two pair of identical digits appear together $= \left(\frac{6!}{2!} - 5! \right) \times 3 = 720$

$$\text{Total numbers} = \frac{8!}{2! 2! 2!} = 5040; \text{ Probability} = \frac{720}{5040} = \frac{1}{7}.$$